Data Analytics Project: IGN Game Reviews Analysis

Dataset Used: ign.csv

This document outlines a structured, end-to-end data analysis project based on game reviews from the IGN dataset. The project is divided into five scenarios, increasing in difficulty from basic data handling to advanced analysis and visualization, ensuring comprehensive skill coverage.

Project Goal: To analyze game review scores, platform popularity, and genre distribution over time using the `ign.csv` dataset.

The **primary libraries/modules** required for this project are:

- **Pandas:** For all data manipulation, filtering, aggregation, and feature engineering tasks.
- **NumPy:** Specifically for numerical calculations like yearly growth (`np.diff`).
- **Visualization Library (e.g., Matplotlib):** For generating all required plots (Line Graphs, Bar Charts, Pie Charts, Stacked Bar Charts).

🎯 Project Coverage

Category	Skills Covered
Data Preprocessing	Data Loading, Missing Value Handling (Mean/Mode), Type Conversion, Column Removal
Pandas Operations	Grouping, Filtering, Aggregation, Feature Engineering (score_category, editors_choice conversion)
NumPy Calculations	Array conversion, Yearly score growth calculation (np.diff)
Visualization	Line Chart, Bar Chart, Pie Chart, Histogram, Stacked Bar Chart, Saving Plots

Scenario 1: Basic Data Loading & Cleaning (🟢 Easy)

Description:

Now, we are going to read the csv file using pandas and going to display the first and last five rows. Then we are going to perform the basic data cleaning steps like removing unnecessary columns, changing the data type of columns to its original and replacing the missing values. This makes sure that data is ready for the next preprocessing steps.

Task	Description
1.	Load the `ign.csv` dataset using Pandas.
2.	Display the first 5 rows (`head()`), last 5 rows (`tail()`), and the shape of the dataset.
3.	Remove the unnecessary column: `"Unnamed: 0"`.
4.	Check for missing values in `score`, `genre`, and `platform`.
5.	Handle missing values: Fill the numeric column `score` with the mean, and the categorical column `genre` with the mode.
6.	Ensure correct data types: Convert `score` → float and `release_year`, `release_month`, `release_day` → integer.

Results of Scenario 1:

```
Displaying the first 5 rows of data:
   Unnamed: 0 score_phrase  ... release_month release_day
0           0      Amazing  ...             9          12
1           1      Amazing  ...             9          12
2           2        Great  ...             9          12
3           3        Great  ...             9          11
4           4        Great  ...             9          11

[5 rows x 11 columns]
```

Fig – 1.1 Displaying first 5 rows

```
---------------------------------------------------------------------
Displaying the last 5 rows of data:
       Unnamed: 0 score_phrase  ... release_month release_day
18620       18620         Good  ...             6          29
18621       18621      Amazing  ...             6          29
18622       18622     Mediocre  ...             6          28
18623       18623  Masterpiece  ...             6          28
18624       18624  Masterpiece  ...             6          28

[5 rows x 11 columns]
```

Fig 1.2 Displaying last 5 rows

```
------------------------------
The shape of the dataset is :
(18625, 11)
------------------------------
```

Fig 1.3 Shape of dataset

Conclusion:

From Scenario one, We can understand the basic information regarding the dataset size, number of rows and what kind of data it is.

Here as per the ign dataset, we have data regarding types of games, reviews, scores, genre of the game, website links and also about the release year, date and month.

- In this step, I have viewed the data and got to know we have an unnecessary column called Unnamed. So I have removed to make sure the features are optimum and consistent.
- We also have the missing data in three columns likely scores, genre and platform. So I have replaced the NaN values using the statistical values.
- I have used mean of scores to replace NaN's as those are numerical values.
- I have used mode of genre and platform as those are textual kind of data. Mode finds the most occurred frequency of the given data and that value will be stored or replaced in place of null or empty values.
- Then we have also converted the data types according to the data present. This helps us to process the data and plot them into graphs easily.

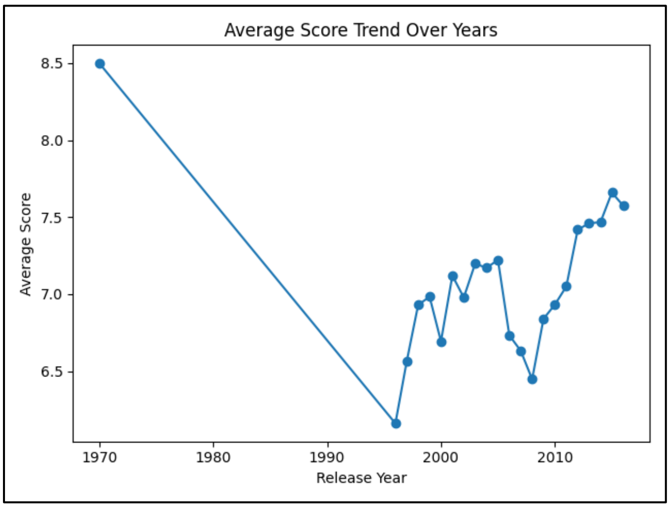
Scenario 2: Line Graph (Score Trend) + Save (▨ Easy)

Description:

Here we are going to construct a line graph showing the basic existing trend line between score over the years. Before constructing the graph we are going to calculate the average score of years so that we can observe whether the trend line is moving upward, downwards or is it in a constant phase

Task	Description
1.	Group data by `release_year`.
2.	Calculate the average score per year using Pandas.
3.	Convert the resulting average scores and years into NumPy arrays.
4.	Plot a line graph with `release_year` on the X-axis and average score on the Y-axis.
5.	Add the title: "Average Game Score Over Years" and proper axis labels.
6.	Save the graph: `plt.savefig("avg_score_trend.png")`.

Graph:



Conclusion:

The graph shows the trend of average game review scores over the years based on data. In 1970, the mean score starts relatively high at around 8.5 but then there is a drastic downfall leading to the 1990s, where the score drops to nearly below 6.5. Late 1990s showed the recovery state steadily indicating improvement in game quality. From mid 1995s to 2000s the score fluctuated moderately showing inconsistency in ratings

After 2010s, the average scores started booming to 7.5 to 7.7 suggesting better overall game reception in recent years. Overall, the graph shows steady improvement in games.

Scenario 3: Filtering + Bar Chart + Save (▨ Medium)

Description:

Now, we are going to analyze the trends a bit deeper by applying filters based on the requirement, We are going to understand the high rated games per platform which gives clear understanding regarding the top rated games. Then visually we will construct a bar chart showcasing only top 10 platforms with high rating games,

Task	Description
1.	Filter the dataset to include only games where `score > 7`.
2.	Count the number of high-rated games per platform.
3.	Select the top 10 platforms using Pandas.
4.	Convert the platform counts into NumPy arrays.
5.	Plot a bar chart with `platform` on the X-axis and count of games on the Y-axis.
6.	Rotate x-axis labels for readability.
7.	Save the graph: `plt.savefig("top_platforms_bar.png")`.

Results for Scenario 3:

```
platform
PC                 1968
Xbox 360            909
PlayStation 2       864
PlayStation 3       783
Wireless            554
iPhone              531
Wii                 512
Xbox                489
PlayStation         424
Nintendo DS         356
Name: title, dtype: int64
```

Fig 3.1 Top 10 Platforms

```
['PC' 'Xbox 360' 'PlayStation 2' 'PlayStation 3' 'Wireless' 'iPhone' 'Wii'
 'Xbox' 'PlayStation' 'Nintendo DS']
[1968  909  864  783  554  531  512  489  424  356]
```

Fig 3.2 NumPy Array of Platforms & Counts

Graph:

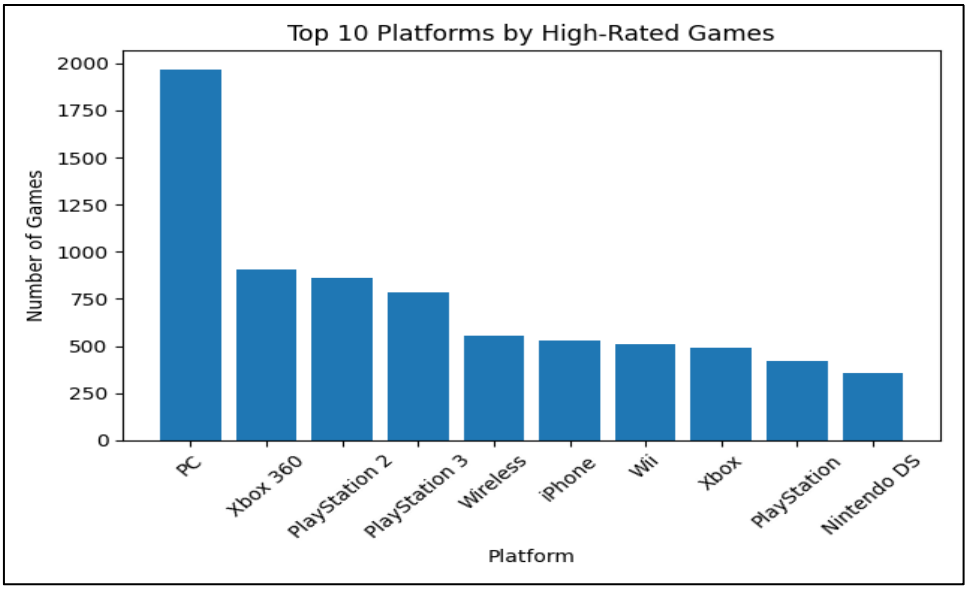


Fig 3.3 Bar Chart of Top 10 Platforms

Conclusion:

The output figure 3.1 represents the top 10 platforms with high rated games which is filtered using scores. And figure 3.2 shows the numpy array where it had platforms and count of each high rated games as values.

Now, from the figure 3.3, the chart represents the top 10 gaming platforms based on the number of high rated games in the IGN dataset.

- PC clearly dominates the list, having the highest number of high rated games far ahead, of all the other platforms.
- Xbox 360 ranks second with a slighter lower count than PC but still higher than the rest.
- Play Station 2 and Play Station 3 follows third position showing strong performance among console platforms.
- Mid range platforms include Wireless, iPhone, and Wii each contributing a moderate number of high rated games
- Nintendo DS has the lowest number of high rated games among the top 10 platforms.
- Finally, the graph highlights that modern and widely used platforms tend to have more high rated games compared to older.

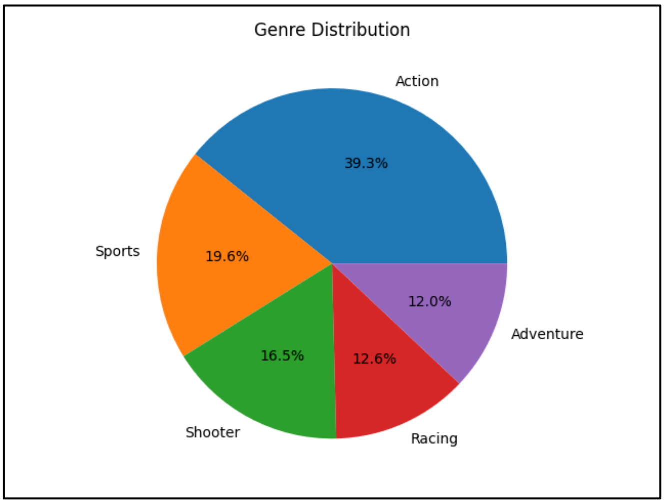
Scenario 4: Aggregation + Pie Chart + Save (🟢 Medium)

Description:

Now we are at the aggregation or data transformation step where it is crucial for deeper analysis. We are going to perform count operation to find total number of games and then we will construct a pie chart followed by genres as labels and count of games as values.

Task	Description
1.	Count the total number of games per `genre`.
2.	Select the top 5 genres using Pandas.
3.	Prepare labels (genres) and values (counts) for the pie chart.
4.	Plot a pie chart using `genre` as labels and `count` as values.
5.	Add percentage labels using `autopct` (e.g., '%.1f%%').
6.	Save the graph: `plt.savefig("genre_distribution.png")`.

Graph:



Conclusion:

The chart shows the distribution of game genres in the data. Action is the most dominant genre, contributing the largest share at 39.3%, indicating its high popularity.

- Sports is the second most common genre, making up 19.6% of the total.
- Shooter games are of 16.5% showing a strong presence among players.
- Racing contributes 12.6% representing a moderate share of the dataset.
- Adventure has the smallest proportion at 12.0% indicating relatively fewer games in this category.
- Finally, the chart highlights that Action games are heavily demanded while other are more evenly distributed but significantly smaller in comparison.

Scenario 5: **Advanced Analysis + Multiple Graphs (● Hard)**

Description:

Now we are performing advanced analysis by creating calculated columns and categorical columns. We have created a categories of Yes and No as 1 and 0 for our usability. And then added a score category column which filters the data according to condition and maps as Excellent, Good and Average based on the scores. Lastly, we will construct a Line, Stacked bar and a Histogram for visualizing trends of each category.

Task	Description
1.	Create a new column `score_category`: "Excellent" for score > 9, "Good" >= 7 score < 9, and "Average" for score < 7.
2.	Convert the `editors_choice` column: 'Y' -> 1$, 'N' -> 0$.
3.	Use NumPy's `np.diff()` to calculate the **yearly score growth** based on the average yearly scores from Scenario 2.
4.	Plot a **Line Graph** showing the trend of average score per `release_year`.
5.	Plot a **Stacked Bar Chart** showing the count of `score_category` per `release_year`.
6.	Plot a **Histogram** showing the distribution of `score`.
7.	Save the graphs: `plt.savefig("score_trend.png")`, `plt.savefig("score_category_stacked.png")`, `plt.savefig("score_distribution.png")`.
8.	Identify: Which years had the highest scores, whether high scores increased over time, and if `editors_choice` correlates with high scores.

Graph:

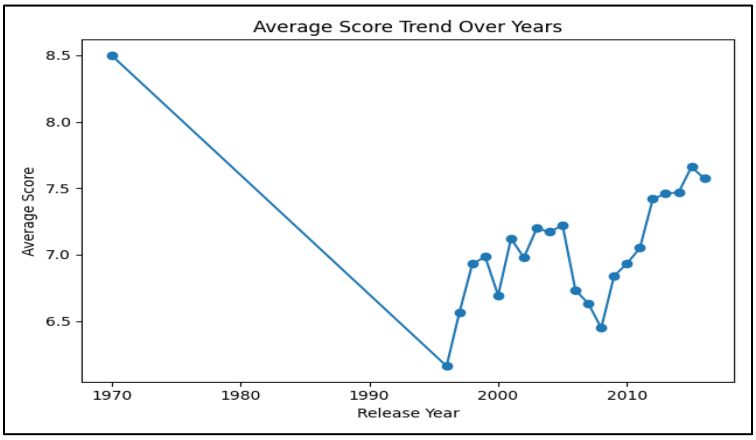


Fig 5.1 Score Trends

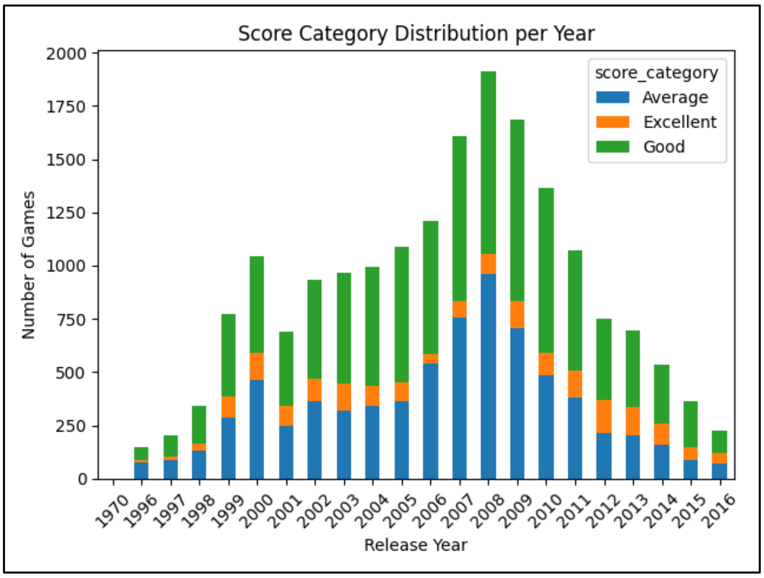


Fig 5.2 Score Category Distribution in Timeline

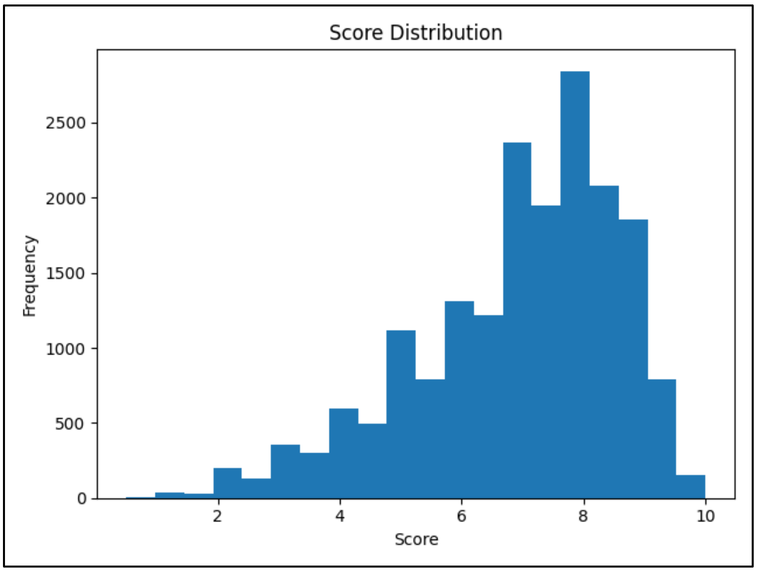


Fig 5.3 Score Distribution in Histogram

Conclusion:

- The line chart shows a sharp decline in average scores from 1970 to mid 1990s, followed by gradual recovery and steady rise after 2000. From 2010, scores stabilized and reaching around 7.5 above in recent years.
- This stacked bar chart shows the yearly count of games categorized as Average, Good and Excellent. The total number of games peaks between 2006 and 2009 indicating a period of high game production
- The distribution is appearing slightly right skewed or a normal like indicating that most games are moderately rated with fewer extreme values.
- All the charts together indicate an increase in game releases over time along with gradual improvement in overall game quality.

Final Conclusion:

Finally, I can conclude that across the analysis, a clear pattern emerges in both game quality and production over time. The average score trend over years shows an initial decline in scores until mid 1990s followed by a steady recovery and improvement in later years. The histogram also shown that most games fall within some range with a peak around 7 to 7.5, it means that the majority of games are moderately to well rated. Extremely high or low scores are relatively rare, suggesting a consistent standard in game reviews.

At the same time, the score category distribution per year highlights an important increase in the number of games released specially between 2005 and 2009, with Average, Good and Excellent categories. The platform chart shows that PC and major consoles contribute the most high rated games, while the genre distribution reveals a strong dominance of Action games. Overall, these insights suggests that the gaming industry has grown in both volume and quality over time, with improvements in development standards and a successful titles on popular platforms and genres.

What you have learned

From this project, I learnt new things like masking the categories into numbers for our usability, adding the columns which will filter whole column data and adding a tags like Excellent, Good and Average for the data based on the condition and many more. The visuals which I have created gave more information than the physical data. Visuals speaks the hidden patterns and trends existing in the data.